

About:
– 2–cell to 7–cell battery managment system designed to handle and protect Li–Ion and Li–Po batteries.

Features:
– Estimates battery state of charge (SoC), state of health (SoH) and others
– Reads battery voltage, cells voltages and battery current
– Reads battery temperature (NTC) and ambient (board) temperature
– Balances battery cells
– Provides I2C and CAN communications to the user
– Powerdown mode consumes < 2uA

Protections – there are 3 levels of protection
– Software protections, MCU control, safely shutting down battery charge/discharge
– Hardware protection (TPS1200 switch) which detect battery overvoltage and overcurrent and shuts it down, independetly of MCU
– Fuse and TVS protection – in case that software and TPS1200 switch fails, there is a fuse which burns and physically disconnects the battery.

Revision B Change Log

Power Control & Protection

Fix – Add manual enable switch to bring the board out of shutdown, rather than relying solely on an external signal holding EN/UVLO high
Fix – Retune EN/UVLO and OV divider resistors to land both trip points at the intended pack voltages, and give the PMOS load–switch gate (Q3) adequate V_{GS} margin across the full input range
Fix – Review GND_BATT connection for voltage measuring paths

Analog Front–End & Measurement Accuracy

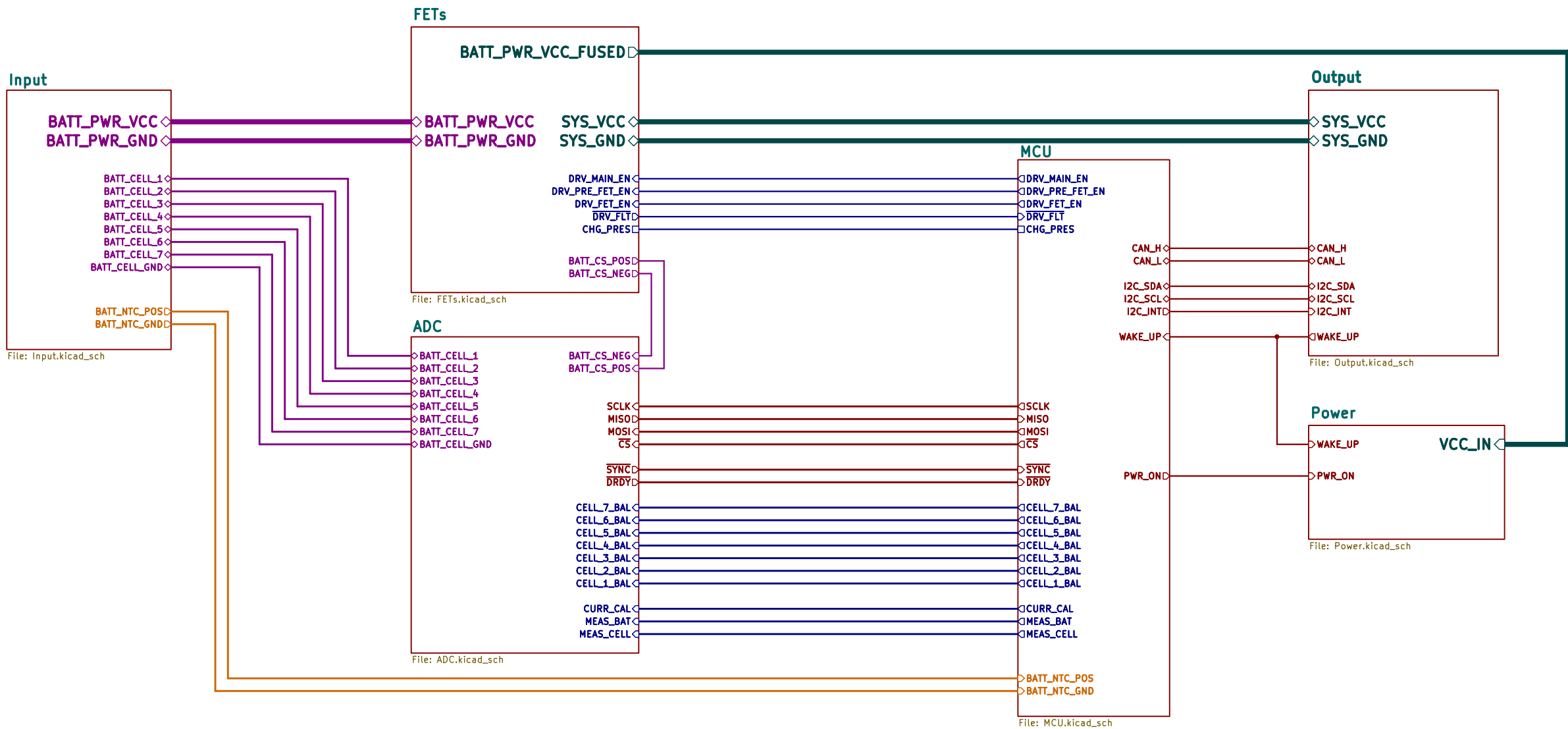
Fix – NTC divider polarity: fixed resistor was wired to GND instead of VCC, invalidating the temperature reading
Fix – Use 0.1% resistors on measurement dividers to reduce channel–to–channel gain mismatch and tighten protection–threshold accuracy
Fix – Measure SYS_VCC to detect charger presence directly, instead of inferring it from current direction
Fix – Add current–sense offset calibration path: switch to short the current–sense differential inputs so offset can be characterized independent of load current
Fix – Reference all measurement dividers to BATT_GND to remove ground–offset error between channels

Component & Schematic Corrections

Fix – LED package change: side–view LED replaced with a top–view package
Fix – Dual NMOS + PMOS schematic symbol didn’t match the physical part’s pinout – corrected
Fix – FET SRC–to–driver connection: driver’s source reference now matches the FET’s actual source node

PCB Layout

Fix – FET footprint–to–zone connection: power FET pads now properly stitched to copper pour for current–carrying capacity and thermal dissipation



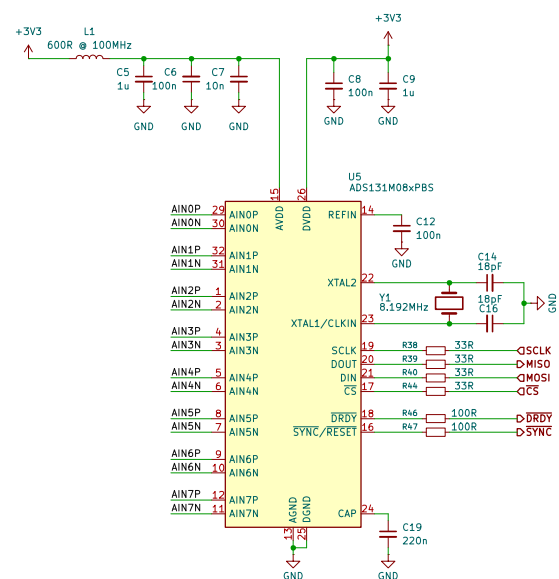
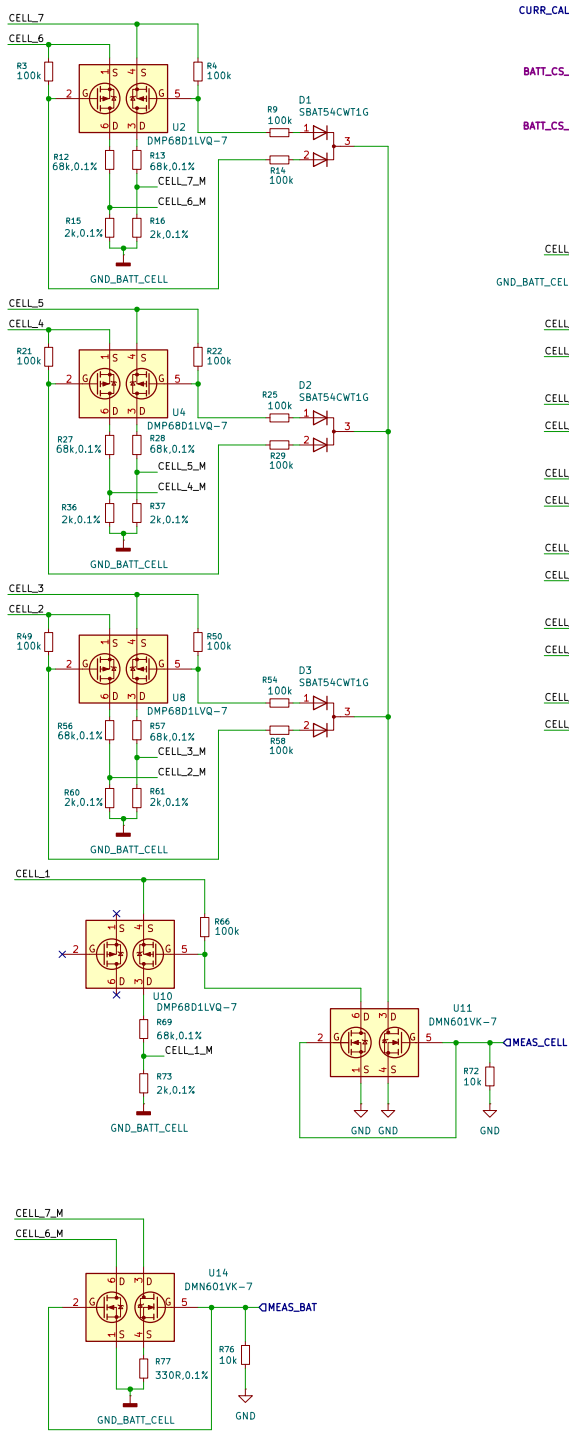
Open Source Battery Management System
Designed by OpenBatt team
More at openbatt.dev

Sheet: /
File: openbms.kicad_sch

Title: OpenBMS

Size: A3 Date: 2026–09–24
KiCad E.D.A. 9.0.7

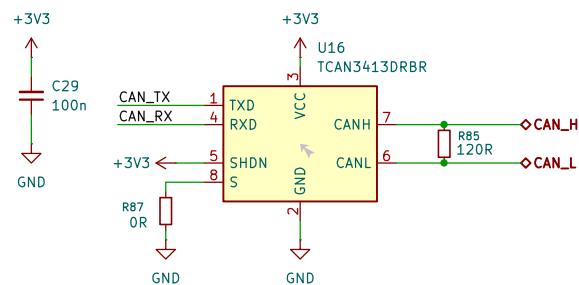
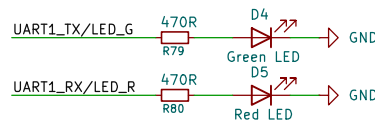
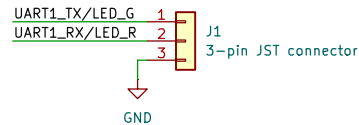
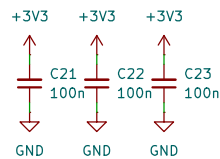
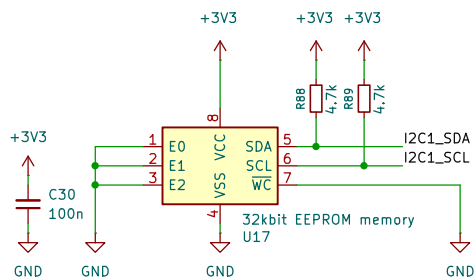
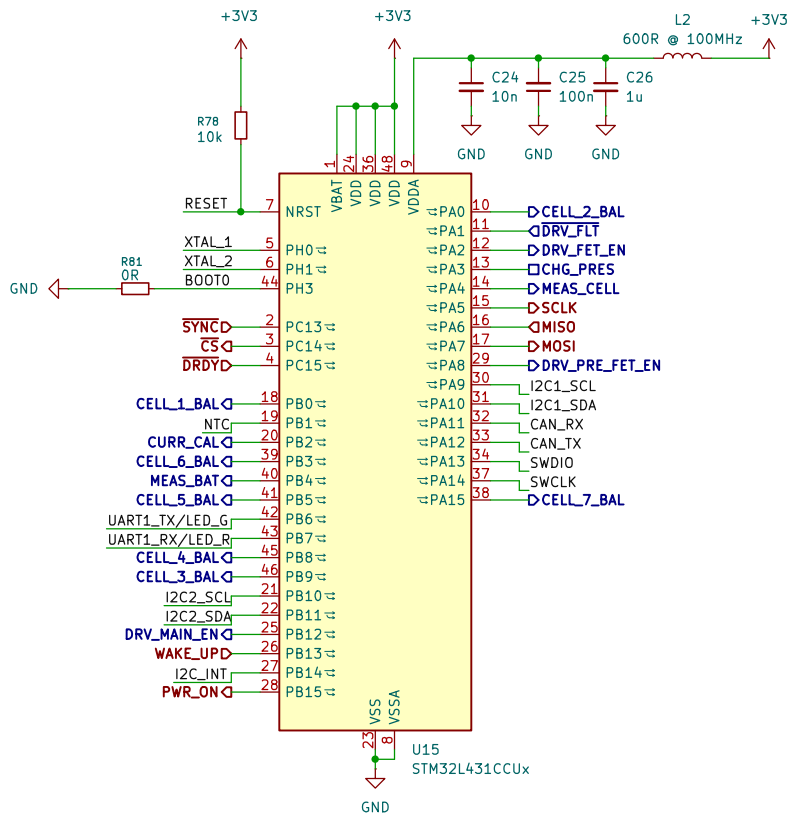
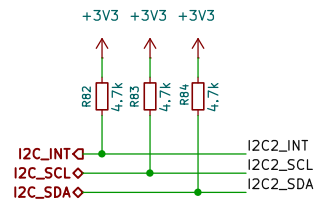
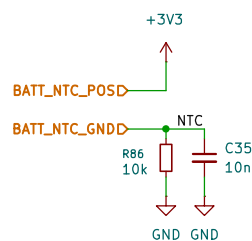
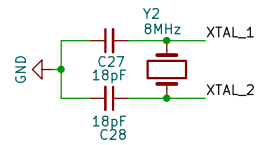
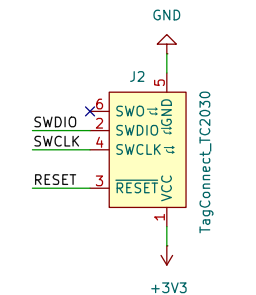
Rev: B
Id: 1/7



Rev: B

About:
- STM32L43 MCU is system brain

Functionalities:
- Configure and reads data from ADC (ADS131M08)
- Controls balancer FETs
- Controls main switch (TPS1200)
- Reads battery NTC
- Stores battery data into EEPROM
- Handles user communication I2C and CAN
- Provides UART debug communication



Open Source Battery Management System
Designed by OpenBatt team
More at openbatt.dev

Sheet: /MCU/
File: MCU.kicad_sch

Title: OpenBMS

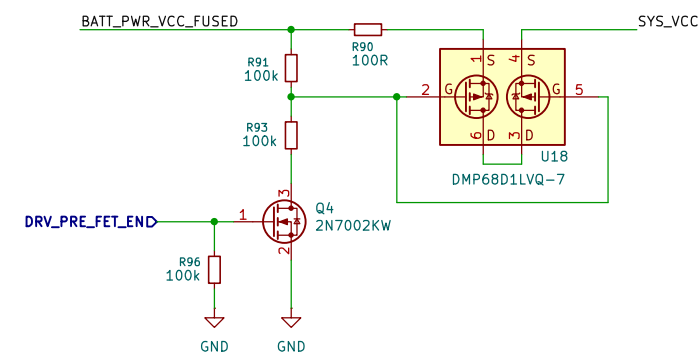
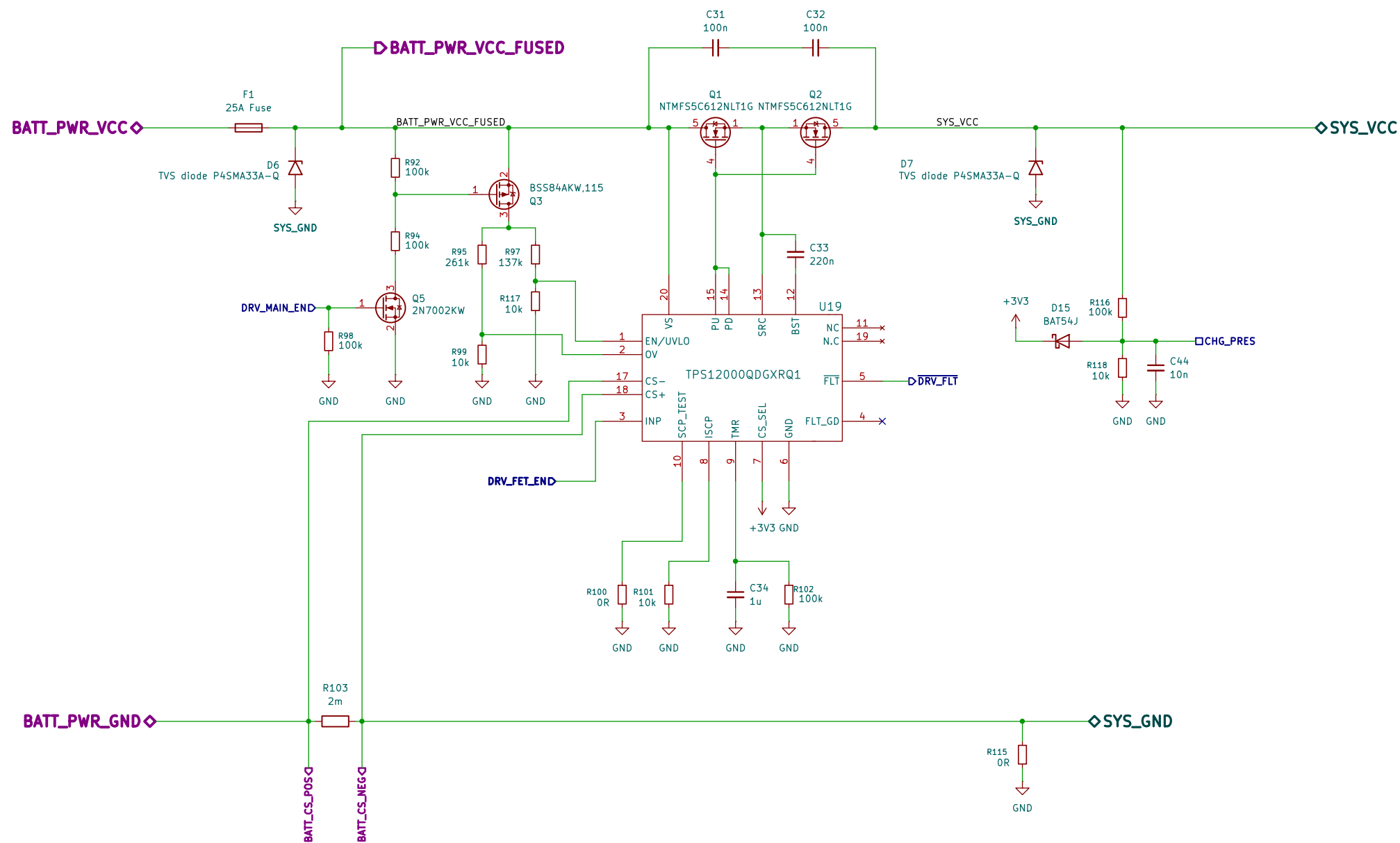
Size: A3 Date: 2026-09-24
KiCad E.D.A. 9.0.7

Rev: B
Id: 3/7

Design Parameters:

- 7-cell battery, minimum cut-off voltage $2.5V \times 7 = 17.5V$ (set by EN/UVLO and OV resistors)
- 7-cell battery, maximum cut-off voltage $4.3V \times 7 = 30.1V$ (set by EN/UVLO and OV resistors)
- Maximum battery cut-off current = 20A (set by shunt resistor and ISCP resistor)
- Burn-out fuse will blow at 25A within 0.3-0.6S

- 1) Set DRV_MAIN_EN to 1 to enable TPS1200
- 2) Set DRV_PRE_FET_EN to 1 to start precharge/predischarge (optional)
- 3) Set DRV_FET_EN to 1 to enable main FETs. This enables battery charge/discharge
- 4) Watch for DRV_FLT_GD. When 1, FET gate voltage is good
- 5) Watch for FLT. When 0, there is a fault present (overvoltage, undervoltage, excessive current..)



Sheet: /FETs/
File: FETs.kicad_sch

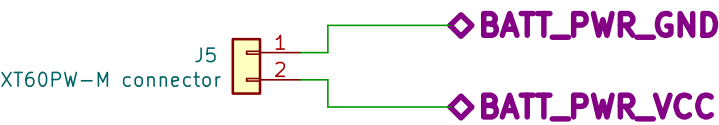
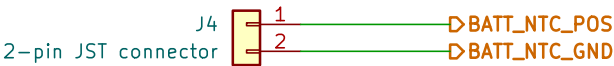
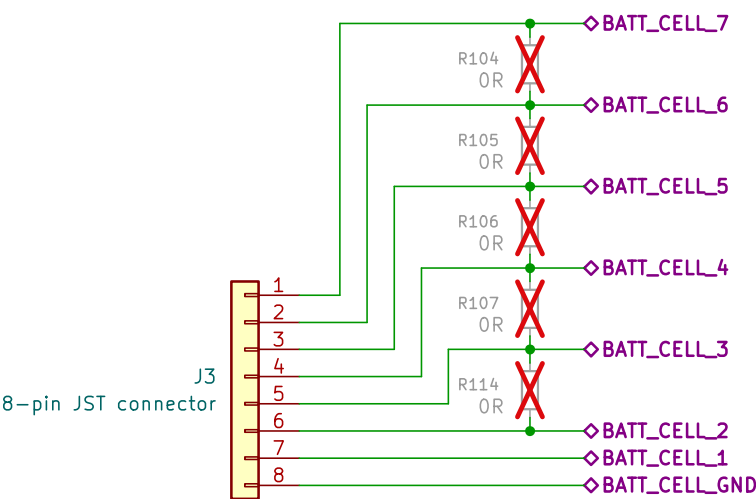
Title: OpenBMS

Size: A3	Date: 2026-09-24
----------	------------------

Size: A3	Date: 2026-09-24	Rev: B
KiCad E.D.A. 9.0.7		Id: 4/7

About:
– Input of the battery. XT60 power connector + 8-pin JST connector for cell wirse + 2-pin JST connector for battery NTC

Design info:
–If battery has less then 7 cells, then place jumper between cell connections which are not being used, e.g. for 5-cell battery jumpers between cells 6 and 7 and betwenn 5 and 6 need to be populated



Open Source Battery Management System
Designed by OpenBatt team
More at openbatt.dev

Sheet: /Input/
File: Input.kicad_sch

Title: OpenBMS

Size: A4

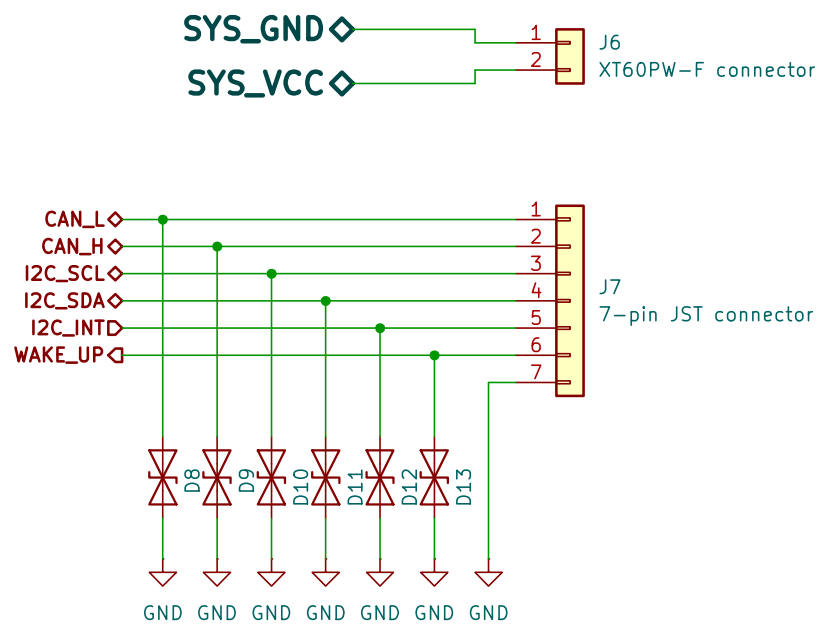
Date: 2026-09-24

Rev: B

KiCad E.D.A. 9.0.7

Id: 5/7

About:
– Output of the BMS, power XT60 connector + JST connector with signals



Open Source Battery Management System
Designed by OpenBatt team
More at openbatt.dev

Sheet: /Output/
File: Output.kicad_sch

Title: OpenBMS

Size: A4

Date: 2026-09-24

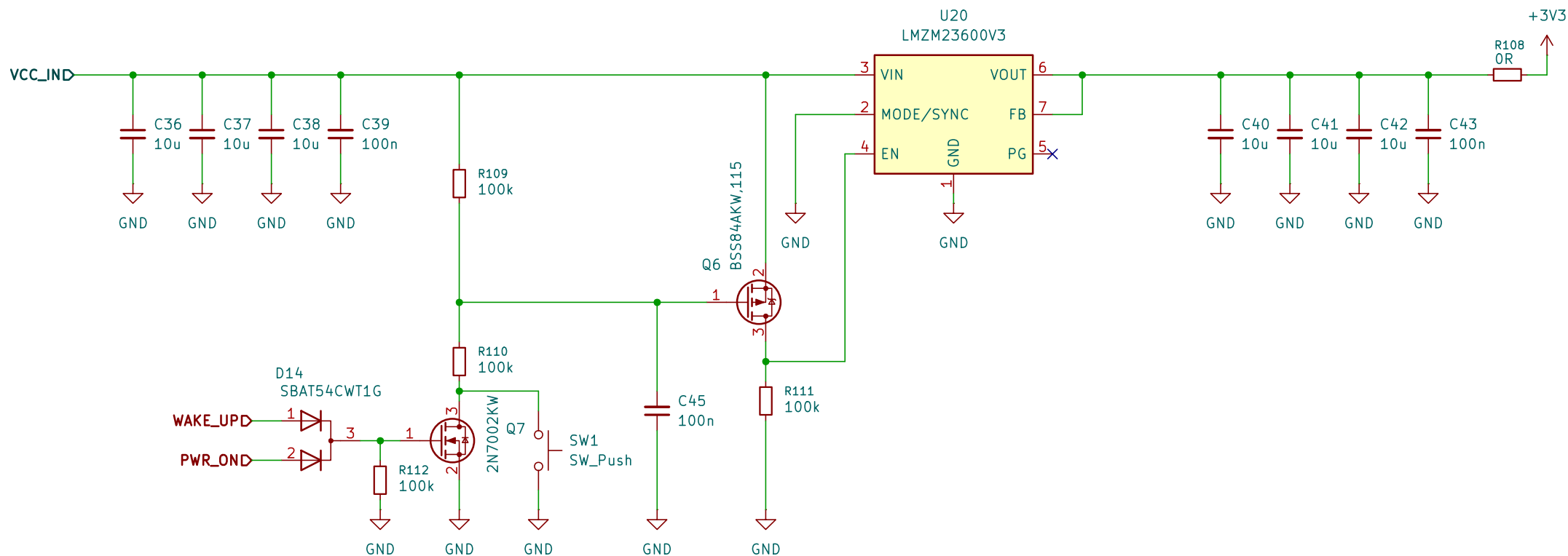
Rev: B

KiCad E.D.A. 9.0.7

Id: 6/7

About:

- Buck converter generates 3.3V out of 4-35V, output up to 500mA.
- By default, power converter is off and whole system too. To enable it, signal from outside WAKE_UP needs to be set to 1. After that, once STM32 controller gets on, it can continue keeping power on by setting PWR_ON signal to 1.
- You can also enable power supply by using on-board switch



Open Source Battery Management System
Designed by OpenBatt team
More at openbatt.dev

Sheet: /Power/
File: Power.kicad_sch

Title: OpenBMS

Size: A4

Date: 2026-09-24

Rev: B

KiCad E.D.A. 9.0.7

Id: 7/7